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Research questions or hypotheses

Primary Research Question

The overarching aim of this review is to examine whether psychometric reporting practices have changed over time across psychology. Our primary and secondary research questions are as follows:

1. Has the overall quality of psychometric reporting for newly developed and modified measures improved over time (2002 to 2026)?

Secondary research questions:

2. Has the type of validity evidence reported (e.g., dimensionality, internal consistency, convergent validity, discriminant validity, relations with other variables) changed over time?

3. Have specific development procedures (e.g., expert review, cognitive interviewing, piloting) of item generation changed overtime?

Foreknowledge of data or evidence

Analyses in this plan have been conducted already. At least some of the analyses described in this analysis plan have been conducted by the authors making this a retrospective registration.

Explanation of foreknowledge and managing unintended influences

This registration reproduces a registration draft that was prepared before the analysis was conducted but, due to a technical error on the submitter's part, was not successfully submitted at that time. The present registration was therefore created after the analysis had been carried out, and is retrospective in timing, although the analysis plan it describes was specified in advance and has not been altered in light of the results.

The following points clarify the extent of foreknowledge and the steps taken to limit its influence. The complete analysis plan, including the outcome definition, the multilevel Bayesian beta regression, the piecewise structure with a fixed 2014 breakpoint, the prior specifications, and the



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regression, the piecewise structure with a fixed 2014 breakpoint, the prior specifications, and the directional hypotheses, was fixed in the original unsubmitted draft prior to conducting the analysis, and is transcribed here without modification. No aspect of the analysis strategy was chosen or adjusted after the results were known. The coding scheme and scoring rules were likewise fixed in advance and are unchanged.

The original draft, with its earlier timestamp, is available and can be provided to document that the plan predates the analysis. The primary and secondary hypotheses, the model structure, the priors, and the decision rules for interpreting the estimates were all specified before any model was fitted.

Readers should nonetheless weight this registration as retrospective in submission, and interpret it accordingly. The analysis plan was preregistered in intent and content but not in successful submission, and this limitation is disclosed here in full.

Data collection is still ongoing and only a part of the whole data has been accessed.

Research Design

Study type

Systematic review: A meta-analysis or other form of systematic review of existing research including scoping reviews.

Descriptive study: Describing some features of a dataset or sample, but typically not for the purposes of informing a causal effect.

Intention for causal interpretation

Blinding of experimental treatments

No blinding is involved.

Additional blinding during research or analysis

No data

Study design

This study employs an observational, cross-temporal content-analysis design. It involves no experimental manipulation, no random assignment of conditions, and no treatment administered to participants; the units of analysis are published journal articles and the psychometric measures reported within them, rather than human participants. The predictor of primary interest is the publication year of each article, treated as a continuous variable spanning



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Study design

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No files selected

Randomization

No data

Sampling

Data collection procedures

All journal article data will be drawn randomly from the open-source OpenAlex database. OpenAlex has good coverage of academic articles, with a database several orders of magnitude larger than that of Web of Science and Scopus (Culbert et al., 2025), and provides a free API supporting bulk extraction, detailed filtering, and randomisation. Data on journal and publication metrics will be drawn from Clarivate's Journal Citation Reports (JCR) database. To avoid inadvertently excluding interdisciplinary research published in non-psychology journals, records will be drawn from both the Psychiatry/Psychology and Social Sciences, General JCR category groups, which together comprised 5,969 journals at the time of pre-registration.

Data will be filtered against two sets of inclusion and exclusion criteria, one relating to paper-level metadata and one to journal-level metadata. Paper-level data will be retrieved and filtered first during the initial API call, with the associated journal-level data filtered subsequently. At the API stage, retrieved articles must have been published between 2002 and 2026, have a DOI and an ISSN or eISSN, not have been retracted, be indexed as belonging to a subfield of psychology, be indexed as type "article," and be written in English. Following the API call, duplicate



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been retracted, be indexed as belonging to a subfield of psychology, be indexed as type "article," and be written in English. Following the API call, duplicate articles will be removed, and the ISSNs and eISSNs of the remaining articles will be cross-referenced against the compiled list of JCR journals; any article whose identifiers fail to match a listed journal will be excluded.

Most filtering is embedded in the API call, but coders will further exclude any article they code that does not contain a newly generated or modified psychometric measure, or that explicitly states that all relevant measures are not intended for research use. Because coders have access to full-text papers, bibliographic fields cannot be obscured; to reduce bias, any coder who reports involvement in the conception, design, analysis, or writing of an assigned article, or any other conflict of interest, will be reassigned to a different randomly selected article.

Owing to the breadth of the inclusion criteria, well over one million papers fall within scope. As manual filtering of all eligible papers is impractical, a randomised subset of 10,000 papers will be drawn for each of five psychology subfields (clinical, applied, cognitive, developmental, and social) using the OpenAlex API, yielding a starting sample of 50,000 articles prior to the removal of duplicates and articles without matching JCR identifiers. Not all articles in the filtered dataset will be coded, as this number will still run to tens of thousands. The minimum sample size for fully coded articles is approximately 1,000, conservatively mirroring sample sizes used in previous studies with similar methods (Barry et al., 2013). As no frequentist analyses are planned, this is not a prescribed stopping point; data collection will continue until the end of the data collection period, which may be extended in the unlikely event that the approximate 1,000-article minimum is not reached within that time.

Data collection procedures - File upload

No files selected

Sample size

This study has no experimental conditions, treatments, or clusters, so sample sizes do not vary across arms of a design. The relevant sampling occurs at two nested levels, and the analysis is multilevel accordingly.

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At the identification stage, a randomised subset of 10,000 papers will be drawn from each of five psychology subfields (clinical, applied, cognitive, developmental, and social) using the OpenAlex API, producing a starting sample of 50,000 articles prior to the removal of duplicates and articles whose identifiers do not match a JCR-listed journal. Not all articles surviving automated filtering will be coded, as this number will still run to tens of thousands.

The primary units of analysis are the individual psychometric measures reported within articles, which form the lower level of the nesting, and the articles within which those measures are reported, which form the upper level. Because a single article may report more than one measure, measures are nested within articles, and the number of measures per article is not fixed but varies across articles. The multilevel model accounts for this dependency through a random intercept for article.

The minimum target for fully coded articles is approximately 1,000, which conservatively mirrors the sample sizes used in previous studies employing similar data collection methods (Barry et al., 2013). This target refers to the article level; the corresponding number of measures at the lower level will be equal to or greater than the number of articles, since each coded article contributes at least one measure and many contribute several. Because no frequentist analyses are planned, this figure is a minimum target rather than a prescribed stopping rule, and data collection will continue until the end of the data collection period, which may be extended if the approximate 1,000-article minimum is not reached within that time.

Sample size rationale

No data

Starting and stopping rules

There is no separate pilot-testing phase that concludes before study data collection begins. Instead, coder calibration serves the function of establishing readiness. Prior to data collection, all coders will complete a pre-selected set of 10

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Starting and stopping rules

There is no separate pilot-testing phase that concludes before study data collection begins. Instead, coder calibration serves the function of establishing readiness. Prior to data collection, all coders will complete a pre-selected set of 10 calibration papers so that intercoder reliability can be assessed. Coding of study data begins once coders have completed calibration and met the reliability criteria. Intercoder reliability will additionally be assessed at the mid-point and end of data collection, using three pre-screened papers at each point. Reliability will be quantified using both Krippendorff's alpha and percent agreement, with acceptable thresholds set at an alpha of at least 0.67 and percent agreement of at least 75% (Krippendorff, 2004). Any coder falling below both thresholds will pause coding, attend a meeting to discuss coding differences, and complete a new set of calibration papers, and must meet the reliability criteria before resuming.

Because no frequentist analyses are planned, there is no power-based or interim-test stopping rule, and no analysis of the outcome will be used to decide when to stop. Data collection will continue until the end of the predetermined data collection period. A minimum target of approximately 1,000 fully coded articles has been set, conservatively mirroring the sample sizes used in previous studies employing similar methods (Barry et al., 2013); this is a minimum target rather than a stopping point. In the event that this minimum is not reached within the data collection period, the period may be extended. Data collection terminates at the close of the data collection period rather than upon reaching any particular result or test outcome.

Variables

Manipulated variables

None

No files selected

Measured variables

Outcome variables. The primary outcome is the overall quality of psychometric reporting for each measure, operationalised as the proportion of applicable coding criteria met. This is computed by scoring each measure against a structured scheme and dividing the number of criteria met by the number applicable, yielding a value on the unit interval. Four secondary outcomes correspond to the





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Outcome variables. The primary outcome is the overall quality of psychometric reporting for each measure, operationalised as the proportion of applicable coding criteria met. This is computed by scoring each measure against a structured coding scheme and dividing the number of criteria met by the number applicable, yielding a value on the unit interval. Four secondary outcomes correspond to the sub-dimensions of the coding scheme, each computed as a proportion in the same way: the reporting of scale construction, the reporting of the sample used in measure development, the reporting of validity and reliability evidence, and the reporting of dimensionality analyses.

Predictor. The primary predictor is the publication year of each article, an observed continuous variable spanning 2002 to 2026. Publication year is entered into the model through a piecewise linear specification, comprising a linear term for the period up to a fixed 2014 breakpoint and a term capturing any change in slope from 2014 onward.

Grouping variable. Article identity, indexed by DOI, is a grouping variable used to model the nesting of measures within articles through a random intercept. It is not a substantive predictor but is required to represent the hierarchical structure of the data.

Measured variables - File upload

No files selected

Indices

Many items can be distilled to a yes or no answer with no distinction between whether the information is in the main article or a supplement. In general, items and sub-items will be scored as 1 (yes) or 0 (no), with variants to this noted below.

Descriptive information about scale construction: Questions 7, 8, 9, 10, and 11 relate to descriptions of how the scale development. For Q9 where there are multiple items under 'yes' these will be scored as 1 (selected) and 0 (not selected). Q10.5 will be scored as (2) 'All items ..', (1) 'Example items ...' and (0) 'No items ...'. This will result in a total count from 0 (reported nothing) to 14 (reported everything that could feasibly be reported)1.

Participant descriptions: Q12-Q14 relate to participant descriptions/missing data. Q12 has three sub-items scored in the normal way resulting (score



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Measured variables - File upload

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Participant descriptions: Q12-Q14 relate to participant descriptions/missing data. Q12 has three sub-items scored in the normal way resulting (score range 0-3); Q13.2 scored treating location, administration and mode as sub-items so the range of scores will be 0-4. Q14 scored in the usual way. Potential range across these items is 0 (reported nothing) to 8 (reported everything).

Validity: Q15.4 and Q15.5 each scored as 1 (they reported something) yielding a total ranging from 0 (not reported) to 2 (reported both convergent and discriminant).

Reliability: Q15.3 scored as 1 for every item selected yielding a total ranging from 0 (nothing reported) to 4 (reported everything)²

Dimensionality: Different scores will be computed for EFA/PCA and CFA

FA/PC Dimensionality: If an EFA or PCA is reported then Q15.1 or Q15.3 items are summed (score 1 for everything selected) to yield a score from 0 to 10 (or more if multiple items are written in other)

CFA Dimensionality: For CFA Q15.2 items are summed (score 1 for everything selected) to yield a score from 0 to 6 (or more if multiple items are written in other)

Overall score (primary outcome): to compute an overall score, we will sum the aforementioned scores.

Indices - File upload

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Analysis Plan



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Analysis Plan

Statistical models

Model type. All research questions are addressed with a Bayesian multilevel beta regression, fitted using the brms interface to Stan. The beta likelihood is used because the outcome is a proportion bounded on the unit interval. Where an outcome contains proportions of exactly 0 or 1, which the beta likelihood cannot accommodate, a zero-and-one-inflated beta model is fitted instead, adding parameters for the number of scores at the boundaries and the proportion of these that equal 1.

Multilevel structure. Because a single article may report several measures, measures (level 1) are nested within articles (level 2). This dependency is modelled through a random intercept for article, allowing each article its own baseline reporting level.

Model specification. The linear predictor is modelled on the logit scale as a piecewise linear function of publication year with a fixed breakpoint at 2014:

$$g(\mu_{ij}) = \ln(\mu_{ij} / (1 - \mu_{ij})) = \eta_{ij}$$

$$\eta_{ij} = [\beta_0 + \beta_1 \cdot \text{time}_i + \beta_2 \cdot (\text{time}_i - 12) \cdot \text{era}_i] + [u_{0j} + \epsilon_{ij}]$$

where μ_{ij} is the expected reporting proportion for measure i in article j ; time is coded as years since 2002 (2002 = 0); era is a dummy variable coded 0 for articles published before 2014 and 1 for 2014 onward; and 12 corresponds to 2014 on the centred time scale. β_0 is the mean reporting proportion in 2002 in log-odds; β_1 is the pre-2014 slope; β_2 is the change in slope at 2014, such that the post-2014 slope is $\beta_1 + \beta_2$; u_{0j} is the article-level random intercept; and ϵ_{ij} is the residual. The two linear segments are constrained to meet at 2014.

Mapping to research questions. The primary research question (whether overall reporting quality changed) is tested with this model fitted to the overall reporting proportion, evaluating β_1 (pre-2014 trend), β_2 (change at the breakpoint), and $\beta_1 + \beta_2$ (post-2014 slope). The two secondary research questions (whether the reporting of specific validity evidence types and of development procedures changed) are tested by fitting the same model separately to each of the four sub-dimensional outcomes: scale construction, sample description, validity and reliability, and dimensionality. The dimensionality model is fitted only to measures reporting a factor analytic procedure.

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evidence types and/or development procedures changed) are tested by fitting the same model separately to each of the four sub-dimension outcomes: scale construction, sample description, validity and reliability, and dimensionality. The dimensionality model is fitted only to measures reporting a factor analytic procedure.

Interactions and contrasts. The only interaction in the model is that between time and era, which constitutes the piecewise breakpoint term. No subgroup analyses, pairwise contrasts, or omnibus follow-up tests are planned. The sub-dimension analyses are parallel applications of the same model rather than contrasts between conditions.

Priors. Priors are specified on the log-odds scale. For the primary model, the slope coefficients are assigned weakly informative priors that are mildly directional, reflecting the pre-registered expectation of improvement, while permitting modest decreases. The intercept is assigned a weakly informative prior spanning plausible base rates. The between-article random-intercept standard deviation is assigned a noninformative prior restricted to positive values. For the secondary models, no directional predictions are made, so their slope coefficients are assigned weaker, non-directional priors. Default weakly informative priors are retained for the dispersion parameter.

Assumption and convergence checks. Prior to fitting, a prior predictive check will confirm that the priors imply plausible values of the outcome. Model convergence will be assessed using the R-hat diagnostic, with values at or below approximately 1.01 taken to indicate convergence, alongside bulk and tail effective sample sizes, with values comfortably above 1,000 taken to indicate adequate sampling. Convergence will additionally be inspected visually through trace plots. Model adequacy will be assessed through posterior predictive checks comparing the observed outcome distribution against distributions simulated from the fitted model. Where a model fails to converge at the initial settings, the number of iterations will be increased and the sampler settings adjusted before the estimates are interpreted; estimates from a non-converged model will not be interpreted.

Statistical models - File upload

No files selected

Transformations

The coded responses are first transformed outcome variables. Within the coding scheme, most items are recoded as binary indicators, scored 1 where the relevant information is reported and 0

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Transformations

The coded responses are first transformed into the outcome variables. Within the coding scheme, most items are recoded as binary indicators, scored 1 where the relevant information is reported and 0 where it is not, with no distinction drawn between reporting in the main article and reporting in a supplement. A small number of items are recoded onto an ordered scale where the scheme distinguishes degrees of reporting, for example an item scored 2 where all items are reported, 1 where example items are reported, and 0 where none are reported. The scored items within each sub-dimension are summed, and the resulting totals are combined into an overall score. The primary outcome is then expressed as the proportion of coding-scheme items reported, taking values on the unit interval.

Publication year is recoded for the piecewise model. The variable time is coded such that 2002 is year zero, with subsequent values representing the number of years since 2002. This centring makes the intercept interpretable as the mean proportion of items reported in 2002 (that is, when both time and era equal zero), expressed in log-odds units. A dummy variable era is coded 0 for articles published before 2014 and 1 for articles published in 2014 or later. The change in slope at the breakpoint is represented by the term $\beta_2(\text{time} - 12) \times \text{era}$, in which 12 corresponds to the year 2014 on the centred time scale. Under this parameterisation, β_1 is the slope for time before 2014, and the slope after 2014 is the sum $\beta_1 + \beta_2$, so that β_2 represents the difference in slope after 2014 relative to before. The two linear pieces are constrained to meet at 2014 (time = 12).

The outcome is modelled on the logit scale through the link function of the beta regression, so no transformation of the outcome is applied beyond the link. Model estimates are expressed in log-odds and are back-transformed to the proportion scale for interpretation. Should any measure take a proportion of exactly 0 or 1, which the beta likelihood cannot accommodate, a zero-and-one-inflated beta model will be fitted, adding parameters that model the boundary values separately from the beta-distributed interior.

Inference criteria

Inference criteria

Inference is conducted within a Bayesian framework, so conventional null-hypothesis significance testing, p-values, and fixed significance thresholds are not used. Inferences are drawn from the posterior distributions of the model parameters.

The directional hypotheses concerning the primary outcome will additionally be evaluated using one-sided Bayesian hypothesis tests, since the pre-registered predictions specify a direction: reporting quality is expected to be flat before 2014 and to improve thereafter. Specifically, the hypothesis that the change in slope at 2014 is positive, and the hypothesis that the post-2014 slope itself is positive, will be tested directionally, and the evidence for these directional hypotheses will be quantified. The secondary sub-dimension analyses are not directional, as no specific predictions are made regarding change in individual sub-dimensions, and their parameters will be interpreted using the same posterior summaries without a directional test.

Data inclusion and exclusion

Data inclusion and exclusion operate at two stages: automated filtering during retrieval, and manual exclusion during coding. Because the outcome is a bounded proportion analysed in a Bayesian framework, no observations are excluded as outliers.

At the retrieval stage, articles drawn from the OpenAlex API are included only if they were published between 2002 and 2026, have a DOI and an ISSN or eISSN, have not been retracted, are indexed as belonging to a subfield of psychology, are indexed as type "article," and are written in English. Following retrieval, duplicate articles are removed, and each article's journal ISSN or eISSN is cross-referenced against the compiled list of Journal Citation Reports journals; any article whose identifiers do not match a listed journal is excluded.

At the coding stage, coders further exclude any article that does not contain a newly generated or modified psychometric measure, or that explicitly states that all relevant psychometric measures are not intended for research use. Purely demographic questionnaire items, and pre-existing validated measures used without modification, fall outside the scope of coding and are not treated as eligible measures. To reduce bias, any coder who reports involvement in the conception, design, analysis, or

At the coding stage, coders further exclude any article that does not contain a newly generated or modified psychometric measure, or that explicitly states that all relevant psychometric measures are not intended for research use. Purely demographic questionnaire items, and pre-existing validated measures used without modification, fall outside the scope of coding and are not treated as eligible measures. To reduce bias, any coder who reports involvement in the conception, design, analysis, or writing of an assigned article, or any other conflict of interest, is reassigned to a different randomly selected article rather than coding that article.

No observations are excluded on the basis of their outcome values. Outliers are not removed, because the outcome is a proportion bounded on the unit interval with no values beyond the possible range, and because the Bayesian beta model accommodates the full observed distribution without the distortions that extreme values can introduce under an unbounded model. All measures within all included articles that meet the eligibility criteria contribute to the analysis, with the sole exception that the dimensionality sub-dimension is computed only for measures reporting a factor analytic procedure, as the relevant items do not apply to other measures.

Missing data

Missing data are addressed by analysing all available data, with the models fitted to the observations present rather than through imputation. Several distinct forms of missingness arise in this design, each handled as follows.

At the article level, articles that fail the automated inclusion criteria, or that are excluded by coders for reporting no eligible measure, are not part of the analysed sample and are not treated as missing data to be recovered; they simply fall outside the eligible population.

At the item level, the coding scheme treats the absence of reported information as a substantive value rather than as missing data. Because the outcome is defined as the proportion of applicable criteria that are reported, an item that is not reported contributes a score of zero to that proportion. Non-reporting is therefore the phenomenon under study rather than a gap to be imputed, and no imputation is performed for unreported items.

At the sub-dimension level, the dimensionality outcome is undefined for measures that report no factor analytic procedure, since the relevant items do not apply. These measures are omitted from the dimensionality model only, and contribute fully to the overall outcome and the other sub-dimension models. This is a structural feature of the coding scheme rather than data missing at random.

Where a measure reuses the sample of a preceding measure within the same article, the relevant participant-description information is carried forward from that preceding measure rather than being treated as missing, reflecting that the information was reported once for the shared sample.

The Bayesian multilevel models are estimated on all available observations. The random intercept for article accommodates the varying number of measures contributed by each article, so articles reporting different numbers of measures are handled naturally without any observation being discarded for imbalance.

Other planned analysis

No data

Other

Context and additional information

No data